Outline

Anatomy of a NN

Design choices

- Activation function
- Loss function
- **Output units**
- Architecture

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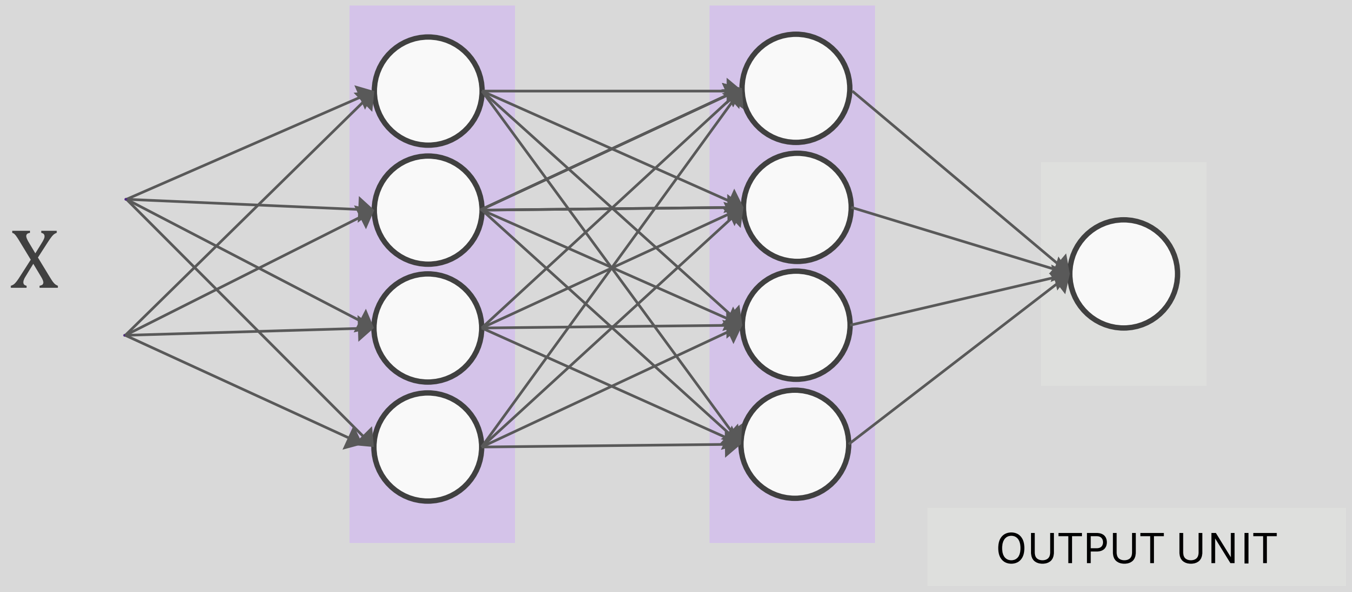
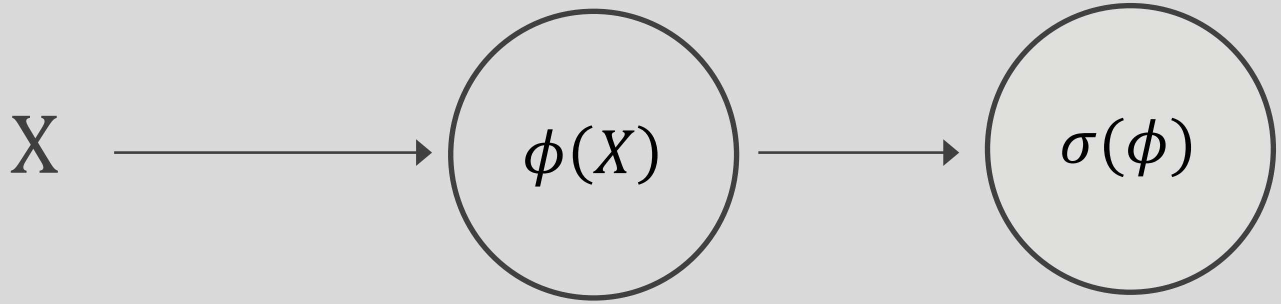
Output Units

Output Type	Output Distribution	Output layer	Cost function
Binary	Bernoulli	?	Binary Cross Entropy

Output unit for binary classification



$\hat{Y} = \mathrm{P}(\mathrm{y} = 1)$



$\hat{Y} = \mathrm{P}(\mathrm{y} = 1)$

$$X \Longrightarrow \phi(X) \Longrightarrow P(y = 1) = \frac{1}{1 + e^{-\phi(X)}}$$

Output Units

Output Type	Output Distribution	Output layer	Cost function
Binary	Bernoulli	Sigmoid	Binary Cross Entropy

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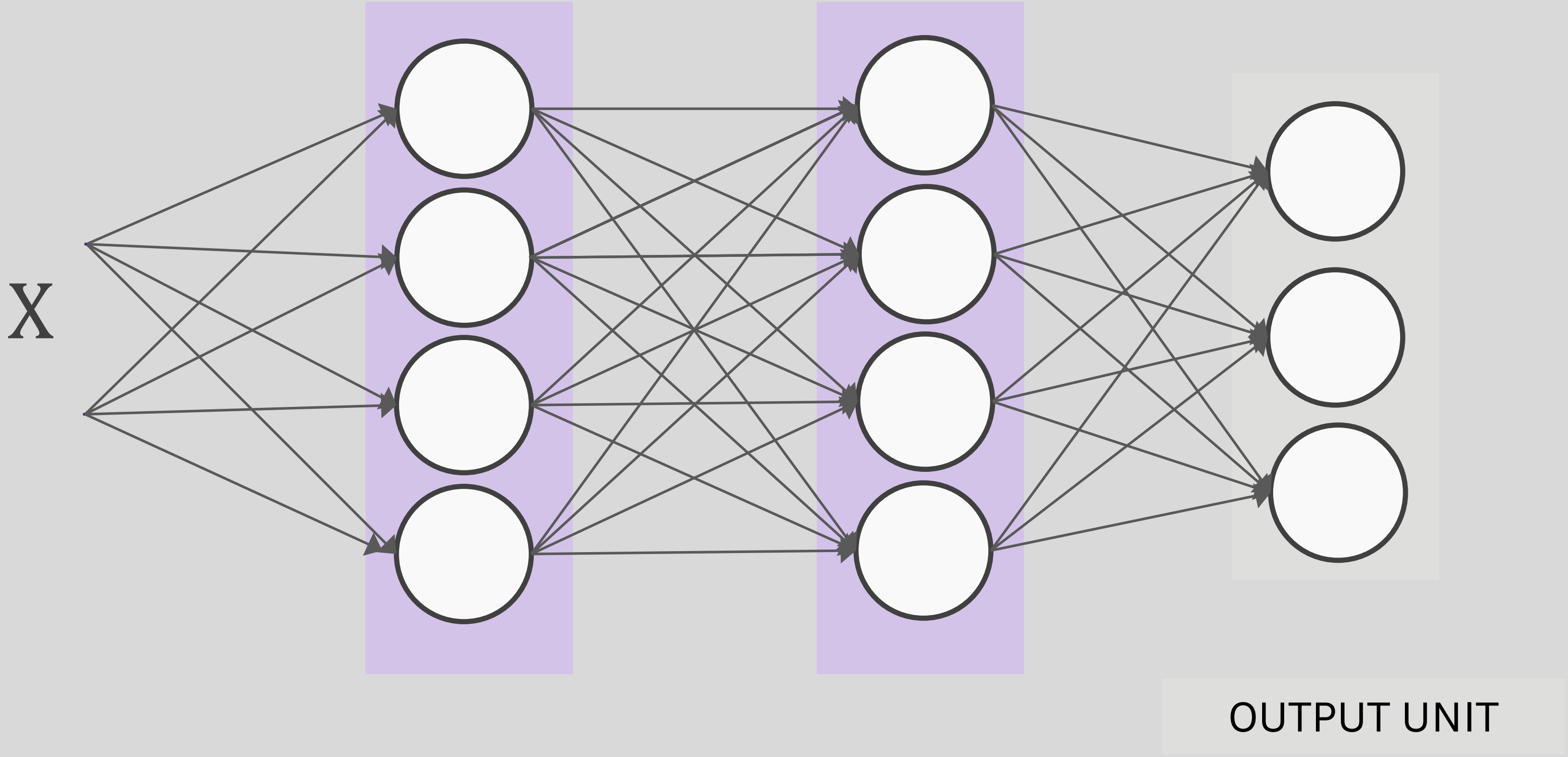
Output Units

Output Type	Output Distribution	Output layer	Cost function
Binary	Bernoulli	Sigmoid	Binary Cross Entropy
Discrete	Multinoulli	?	Cross Entropy

Output unit for multi-class classification

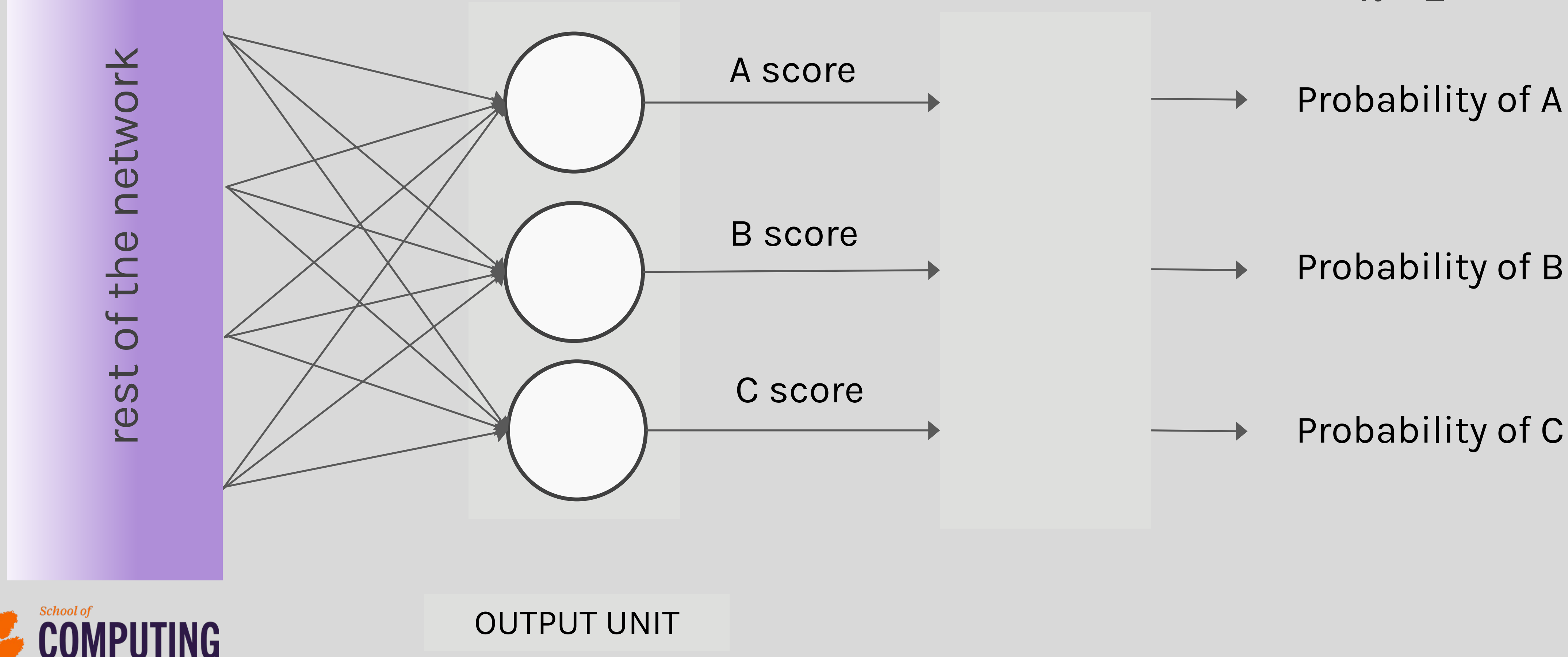


$$\hat{Y} = [P_1, P_2, P_3]$$

SoftMax

$$\phi_k(X)$$

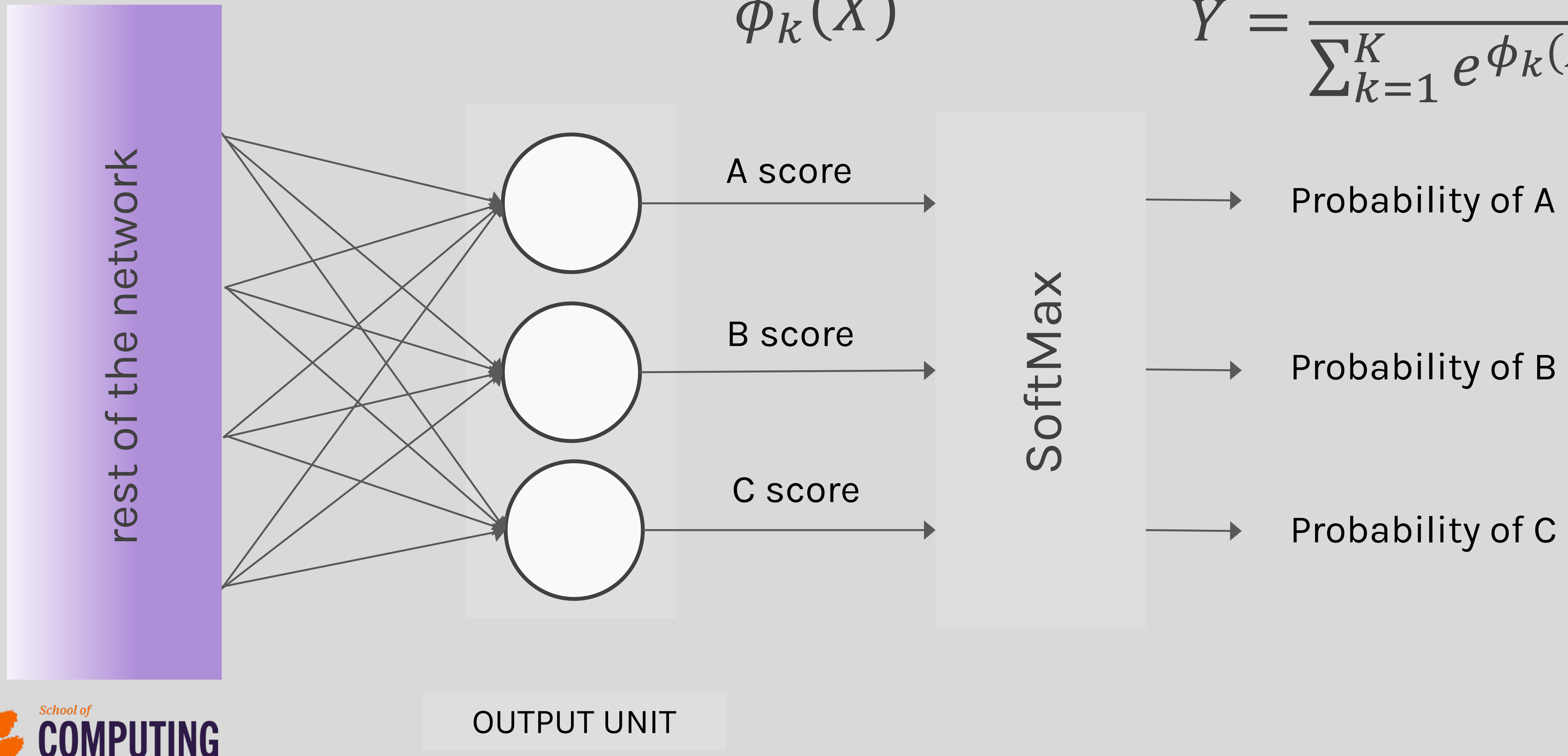
$$\hat{Y} = \frac{e^{\phi_k(X)}}{\sum_{k=1}^{K} e^{\phi_k(X)}}$$



SoftMax

$$\phi_k(X)$$

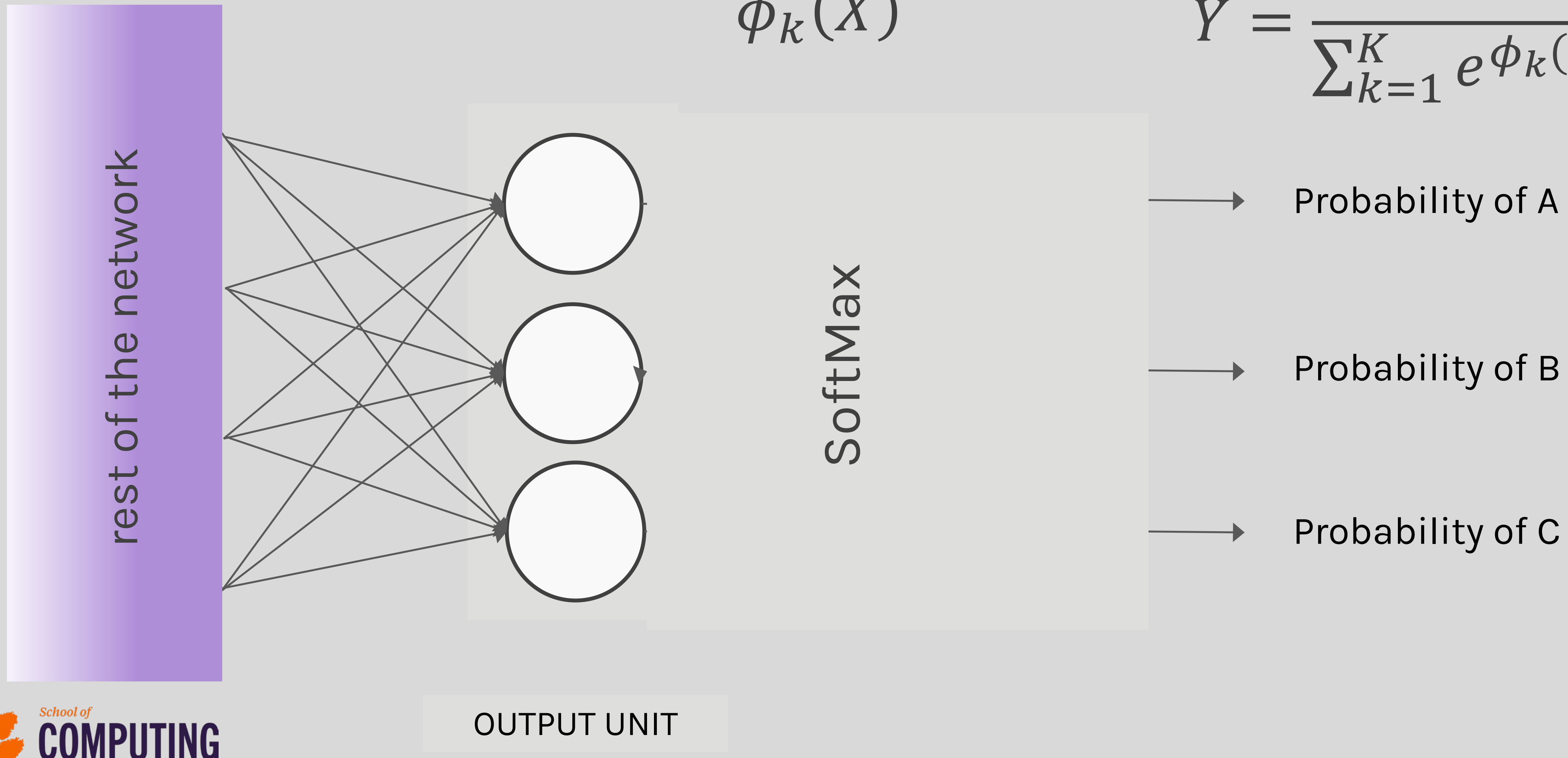
$$\hat{Y} = \frac{e^{\phi_k(X)}}{\sum_{k=1}^{K} e^{\phi_k(X)}}$$



SoftMax

$$\phi_k(X)$$

$$\hat{Y} = \frac{e^{\phi_k(X)}}{\sum_{k=1}^{K} e^{\phi_k(X)}}$$



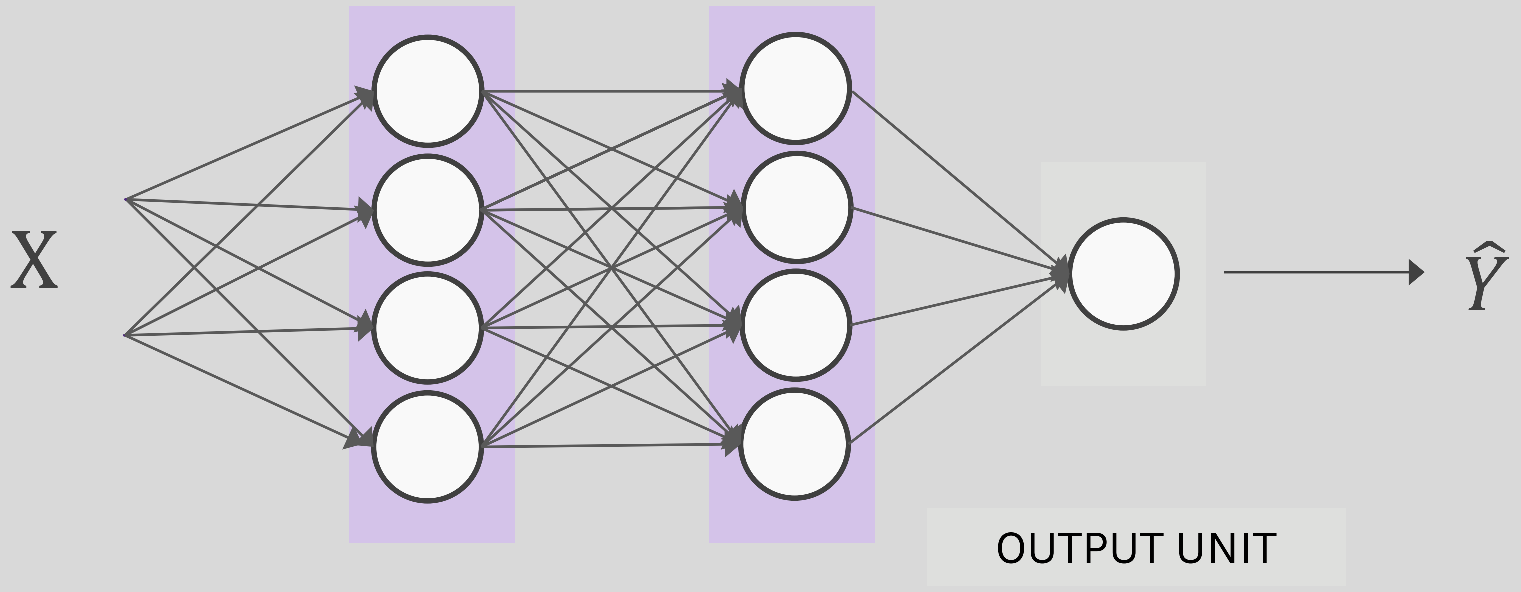
Output Units

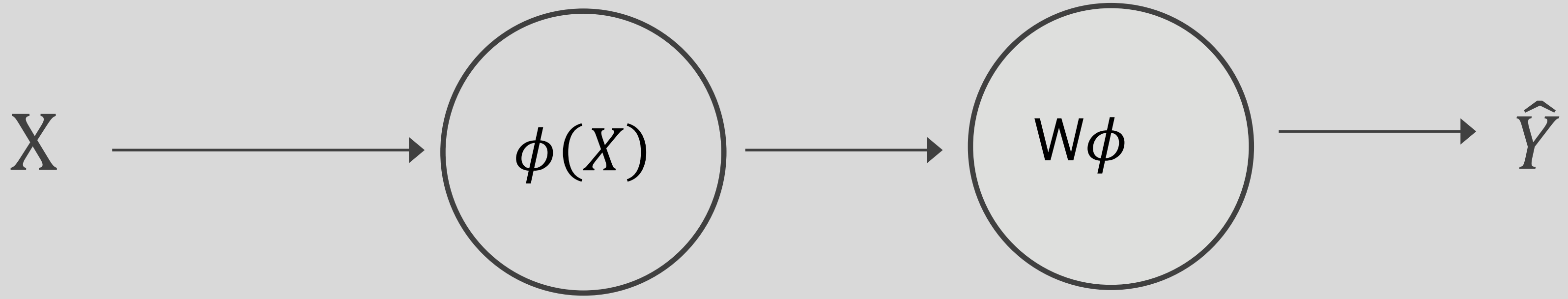
Output Type	Output Distribution	Output layer	Cost function
Binary	Bernoulli	Sigmoid	Binary Cross Entropy
Discrete	Multinoulli	Softmax	Cross Entropy

Output Units

Output Type	Output Distribution	Output layer	Cost function
Binary	Bernoulli	Sigmoid	Binary Cross Entropy
Discrete	Multinoulli	Softmax	Cross Entropy
Continuous	Gaussian	?	MSE

Output unit for regression





$$X \Longrightarrow \phi(X) \Longrightarrow \hat{Y} = W\phi(X)$$

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Output Units

Output Type	Output Distribution	Output layer	Cost function
Binary	Bernoulli	Sigmoid	Binary Cross Entropy
Discrete	Multinoulli	Softmax	Cross Entropy
Continuous	Gaussian	Linear	MSE

Output Units

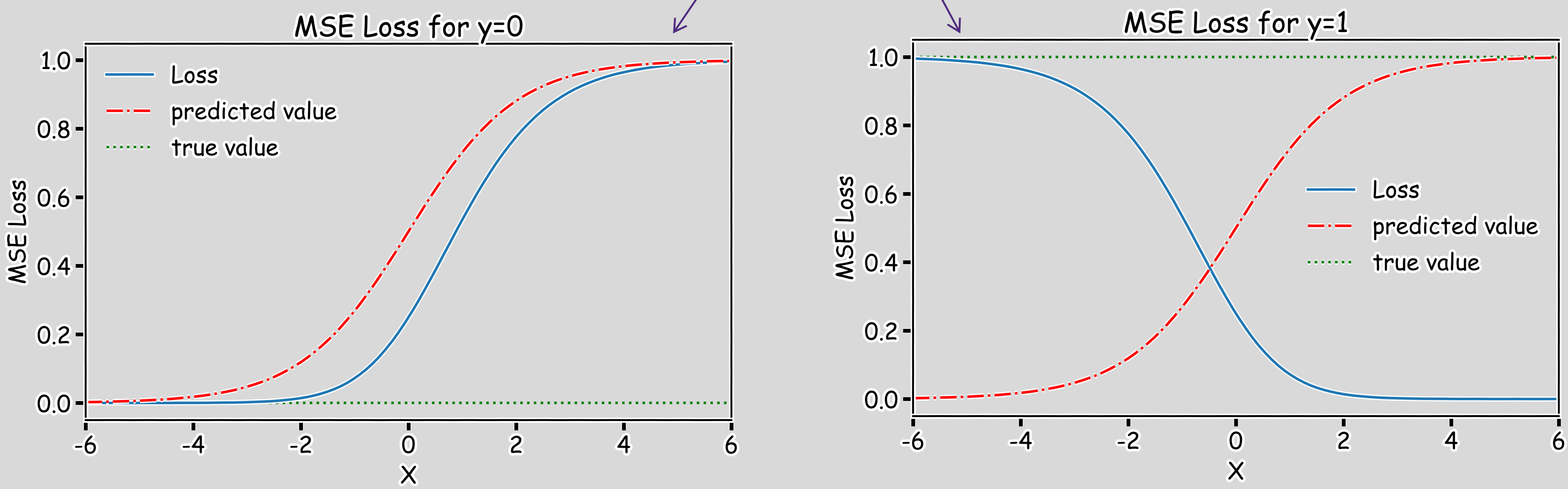
Output Type	Output Distribution	Output layer	Cost function
Binary	Bernoulli	Sigmoid	Binary Cross Entropy
Discrete	Multinoulli	Softmax	Cross Entropy
Continuous	Gaussian	Linear	MSE
Continuous	Arbitrary	-	GANS

Loss Function

Example: sigmoid output + squared loss

$$L_{sq} = (y - \hat{y})^2 = \left(y - \sigma(x)\right)^2$$

Flat surfaces



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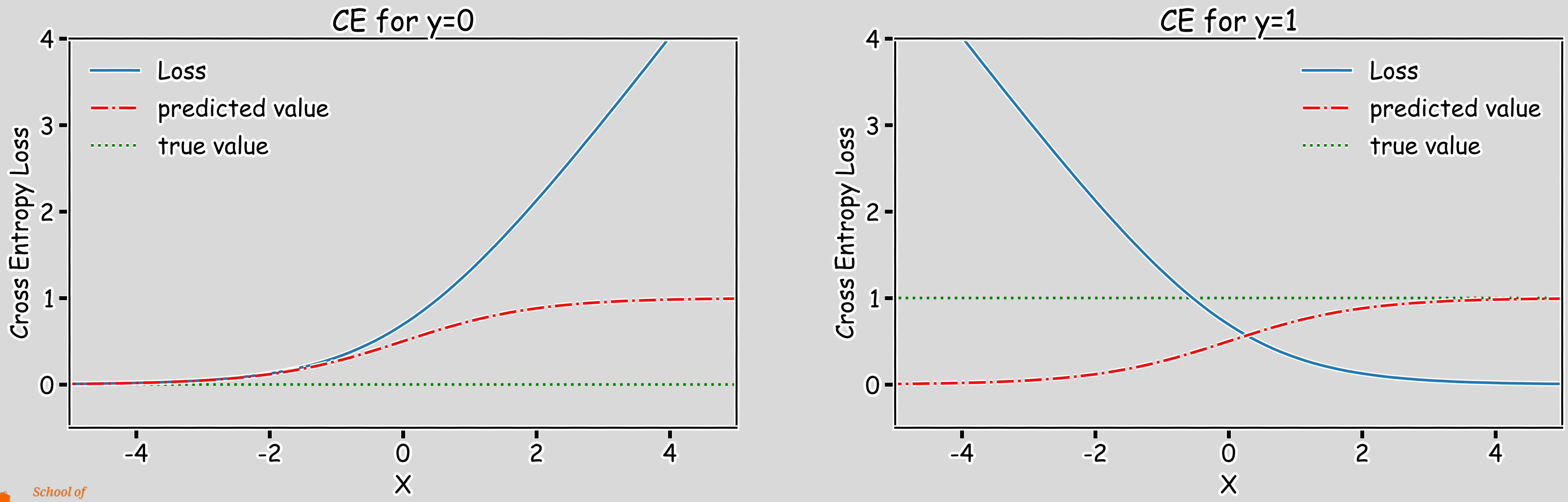
Cost Function

Example: sigmoid output + cross-entropy loss

$$L_{ce}(y, \hat{y}) = -\{\, y \log \hat{y} + (1 - y) \log(1 - \hat{y})\}$$



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